THE DOUBLE TRANSVERSALITY

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Abstract. The preceding paper observed that the collision invariant's Fourier spectrum is anti-correlated with the prime character sum magnitudes at lag 1: large collision coefficients tend to pair with characters having small $|P(s,\chi)|$. This paper reports two computational extensions of that observation, built on the exact transform and neutrality framework from the earlier papers.

First, the anti-correlation persists at every lag. At lag ℓ, the collision invariant lives on $(\mathbb{Z}/b^{\ell+1}\mathbb{Z})^\times$ and involves $\phi(b^{\ell+1})/2$ odd characters. The Pearson correlation between $|\hat{S}^\circ(\chi)|$ and $|P(s,\chi)|$ is negative at every base and lag tested, from 3 characters (base 3, lag 1) to 2,187 characters (base 3, lag 7).

Second, the base sum $F_R(s) = \sum_b w(b)\, F_b^\circ(s)$ nearly vanishes. Individual bases produce $F_b^\circ(0.5)$ values ranging from -1.4 to $+5.1$, but their weighted sum is -0.01. The collision invariant's structural content appears to be base-specific: the individual signals cancel in the aggregate.

The two observations together suggest a double transversality: within each base, the Fourier spectrum is anti-correlated with the zero-sensitive characters; across bases, the resulting signals point in different directions and cancel. The term *double transversality* is used here for that observed pattern, not for a theorem explaining convergence.

1. Introduction

The preceding paper [5] proved the general neutrality theorem and observed a computational anti-correlation: the collision coefficients $|\hat{S}^\circ(\chi)|$ are negatively correlated with the prime character sum magnitudes $|P(s,\chi)|$ across odd characters at lag 1.

This note keeps the status line explicit. The exact backbone comes from the earlier papers: odd-channel selection, centering, and the decomposition of F_b° into odd character sums. The new content here is computational: persistence of the anti-correlation at higher lags, and near-vanishing of a finite weighted base sum across a specified set of prime bases.

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Two questions remained. Does the anti-correlation persist as the lag increases and the family of L-functions grows? And what happens when the collision invariant is aggregated across bases?

2. The Anti-Correlation at Higher Lags

At lag ℓ, the collision invariant S_ℓ lives on $(\mathbb{Z}/m\mathbb{Z})^\times$ with $m = b^{\ell+1}$. The number of odd characters is $\phi(m)/2$, which grows as $b^\ell(b-1)/2$.

b	ℓ	m	odd	$+$	$-$	net	ρ
3	1	9	3	3	0	+0.62	0.00
3	2	27	9	5	4	+1.18	+0.42
3	3	81	27	15	12	+0.62	−0.42
3	4	243	81	35	46	+4.83	−0.28
3	5	729	243	143	100	+12.8	−0.33
3	6	2,187	729	359	370	+29.1	−0.33
3	7	6,561	2,187	1,167	1,020	+11.1	−0.20
5	1	25	10	8	2	+1.04	−0.45
5	2	125	50	34	16	+4.01	−0.18
5	3	625	250	134	116	+2.48	−0.28
5	4	3,125	1,250	636	614	+15.3	−0.19
7	1	49	21	13	8	+1.81	−0.28
7	2	343	147	89	58	+7.56	−0.22
7	3	2,401	1,029	509	520	+4.58	−0.16

Table 1. Anti-correlation at higher lags. The Pearson correlation ρ between $|\hat{S}^\circ|$ and $|P(0.5,\chi)|$ is negative at every lag beyond $\ell = 2$ at base 3. In the tested cases, the net $F^\circ(0.5)$ is positive. 148,000 primes per row.

Three patterns emerge.

The anti-correlation persists at every lag. Beyond the smallest cases (base 3, lags 1 and 2, where there are too few characters for the correlation to stabilize), the Pearson coefficient is consistently negative, ranging from -0.16 to -0.45.

The net $F^\circ(0.5)$ is positive at every base and lag. From 3 characters to 2,187 characters, the sign structure always resolves in favor of convergence.

The net grows with the number of characters. At base 3, it increases from +0.62 (lag 1) to +29.1 (lag 6). The convergence does not weaken as the family of L-functions expands.

3. The Base Sum

Let $\mathcal{B} = \{3, 5, 7, 11, 13, 17, 19, 23, 29, 31\}$ be the set of prime bases used in the experiment. For a fixed prime p, the base sum

$$R(p) = \sum_{b \in \mathcal{B}} w(b)\, S_1^\circ(p, b)$$

aggregates the centered collision deviation across these bases with weight $w(b) = 1/b^2$. The base-summed prime harmonic sum is $F_R(s) = \sum_p R(p)/p^s = \sum_{b \in \mathcal{B}} w(b)\, F_b^\circ(s)$.

s	F_R	$\min F_b^\circ$	$\max F_b^\circ$	range
1.0	-0.000	-0.04	$+0.21$	0.25
0.8	-0.001	-0.15	$+0.78$	0.93
0.6	-0.006	-0.66	$+2.80$	3.46
0.5	-0.010	-1.36	$+5.07$	6.43

Table 2. The base sum $F_R(s)$ compared with the range of individual $F_b^\circ(s)$ across bases $b = 3, 5, 7, 11, 13, 17, 19, 23, 29, 31$. 148,000 primes.

At $s = 0.5$, the individual base values span a range of 6.4, but their weighted sum is -0.01. The base sum nearly vanishes.

b	$F_b^\circ(0.5)$
3	-0.03
5	-0.00
7	$+0.07$
11	-1.36
13	-0.18
17	-0.09
19	-0.55
23	-1.30
29	$+2.80$
31	$+5.07$

Table 3. Individual $F_b^\circ(0.5)$ values. Some bases are positive, some negative. Their weighted sum is -0.01.

4. The Double Transversality

The computations in the preceding sections suggest two levels of cancellation.

Within each base: the centered sum $F_b^\circ(s)$ is a linear combination of character sums $P(s,\chi)$ with mixed-sign products. In the tested cases, the anti-correlation between $|\hat{S}^\circ(\chi)|$ and $|P(s,\chi)|$ accompanies a positive net: the positive terms outweigh the negative at every base and lag tested.

Across bases: the bounded F_b° values have mixed signs across the tested bases, and their weighted sum over $\mathcal{B}$ nearly vanishes.

Definition 1. The *double transversality* is the name for the observed pattern that:

(1) at each base, the collision spectrum is anti-correlated with the prime character sum magnitudes, and
(2) across bases, the individual F_b° values cancel in the weighted sum.

This is a computational observation, not a theorem.

Remark 2. If the double transversality is structural, the collision invariant has base-specific content that cannot be recovered from the aggregate. The convergence at each base would then be a local phenomenon, depending on the digit geometry at that base rather than on a universal property.

5. Interpretation

Remark 3. If the double transversality holds generally, the convergence of $F_b^\circ(s)$ at a specific base b is a local phenomenon, depending on the anti-correlation at that base rather than on a universal property across bases. This would separate the collision transform from approaches that study families of L-functions through averaged quantities: the base sum loses the structural content that is present at each individual base.

Remark 4. The vanishing of F_R does not mean the base sum is uninteresting. The neutrality theorem (the mod-3 splitting of the base sum) is a statement about $R(p)$ itself, not about $F_R(s)$. The double transversality operates at the level of the prime harmonic sum, not at the level of individual primes.

6. Remarks

What is proved, what is computed. The higher-lag anti-correlation and the vanishing base sum are computational observations. The anti-correlation is tested at 3 bases across 15 lag values, from 3 to 2,187 odd characters. The base sum is computed across 10 bases with 148,000 primes.

The general neutrality theorem and the scramble proposition (from the preceding paper [5]) are proved. The double transversality is a name for the observed structure, not a theorem.

What the higher lags reveal. As the lag ℓ increases, the modulus $m = b^{\ell+1}$ grows and the family of odd L-functions expands. The anti-correlation persists throughout the tested range, meaning the collision invariant continues to avoid the zero-sensitive directions in these computations even as the character space grows. This is consistent with the avoidance being structural rather than a small-modulus artifact, but the paper does not prove that conclusion.

The arc. The collision invariant began as a count of digit coincidences [3]. The collision transform revealed its character decomposition and proved convergence at $s = 1$ [4]. The general neutrality theorem constrained the decomposition at every prime scale [5]. The sign structure and anti-correlation suggested that convergence below $s = 1$ may be a collective phenomenon, not a test of individual L-functions [5].

The double transversality extends that picture. The computations suggest the collision invariant's relationship to L-function zeros is structured at two levels: within each base (the spectrum is anti-correlated with zero-sensitive characters) and across bases (the individual signals cancel). Whether this pattern reflects a structural property of the digit function's geometry or an artifact of finite computation remains open.

References

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